

Notes on Accretion Discs

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Abstract

Notes on the Accretion Discs lecture series by Gordon Ogilvie. Used to learn material mainly up to the diffusion equation, not all lectures and example sheet problems have been covered; the example sheet problem solutions are written by me, so may have errors. If you do find any errors or typos, feel free to let me know at davidpuerta-maths@gmail.com

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0 Introduction

0.1 Accretion

Key Points:

- Accretion is the accumulation of particles into an object due to gravity.
- Accretion discs are the above but rotating.

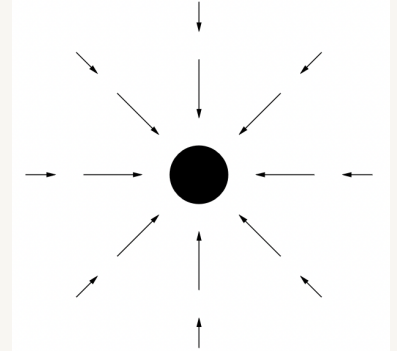
Accretion is the accumulation of particles into an object due to gravity.

Consider a particle of mass m that falls from infinity until it reaches the surface of a star of mass M and radius $R_\star$, whereby it comes to rest. The energy released E , is given by

$$E = \frac{GMm}{R_\star} = \left(\frac{R_S}{2R_\star} \right) mc^2, \quad R_S = \frac{2GM}{c^2}$$

where R_S is the Schwarzschild radius. This inward falling of the particle from infinity to the surface of a star is a form of accretion, and is more efficient than nuclear fusion.

The simplest form of accretion is **spherical/Bondi accretion**. This is best shown by a star in a static, uniform gaseous medium, which accretes mass from its surroundings, as shown in the image to the right. However, a necessary assumption needed for this type of accretion is that the surrounding gas has negligible angular momentum.



Consider a particle of mass m in a circular orbit around a star of mass M in a static, uniform gaseous medium. Suppose that the orbital radius of the particle R can be contracted from its current radius R to a much smaller radius r where $r \ll R$. The particle's kinetic energy in an orbit of radius R can be found using its equation of motion as

$$T = \frac{GMm}{2R}$$

and its potential energy is given by

$$U = -\frac{GMm}{R}$$

both of which give the energy for a particle of mass m in an orbit of radius R as,

$$E = T + U = -\frac{GMm}{2R}$$

Now, the energy released during this transition from an orbital radius of R to r is

$$\Delta E = -\frac{GMm}{2R} + \frac{GMm}{2r} \stackrel{r \ll R}{\approx} \frac{GMm}{2r}$$

This is approximately equal to the [binding energy](#) of the final orbit. To see this, recall that the binding energy is the amount of energy that must be supplied to the particle to remove it completely from the gravitational field, taking it from its orbit at radius r to infinity,

$$E_{\text{bind}} = E_{\infty} - E = \frac{GMm}{2R}$$

where we have take the energy at infinity, E_{∞} , to be zero. However, to achieve this, almost all of the angular momentum of the larger orbit, $m(GMR)^{1/2}$, must be removed. This removal of angular momentum is a key property of accretion discs. Most accretion flows are rapidly rotating. An [accretion disc](#) is a flow that achieves an outward transport of angular momentum.

Consider a static uniform spherical cloud of gas that collapses under gravity (gravitational collapse). The collapse can be treated as spherically symmetric. Due to this collapse, an object forms that is supported against its gravity by pressure.

If the cloud is initially in uniform rotation, its dynamics will be affected by its centrifugal force; this resists collapse in the perpendicular plane to the axis of rotation. Initially negligible, it becomes more important as it collapses, conserving angular momentum.

0.2 Discs in Astrophysics

Key Points:

- Protostellar discs, binary stars, active galactic nuclei and such are examples of such astrophysical accretion discs.

0.2.1 Protoplanetary discs

Planets of the solar system orbit the sun. This is due to the planets condensing out of a flattened cloud of gas rotating around the Sun. [Protostellar discs](#) consist of relatively cool gas, mostly H_2 and dust and survive for a few million years.

0.2.2 Interacting binary stars

These systems consist of two stars: the primary and secondary. The primary evolves more rapidly, whilst the secondary will split off from its companion. The transferred gas has a large

amount of angular momentum and forms an accretion disc around the collapsing (primary) star. The gas gradually loses angular momentum and spirals inwards to the central object.

- White dwarf primaries are called [cataclysmic variables](#) which include classical novae and dwarf novae.
- Black hole primaries are called [X-ray binaries](#) which include low/high-mass X-ray binaries.

0.2.3 Active galactic nuclei

Most galaxies contain a supermassive black hole at the centre. Active galactic nuclei (AGNs) include an accretion disc around the black hole, which provides energy for the hole to grow.

0.2.4 Other astrophysical discs

- [Spiral galaxies](#) are discs of stars and gas, they are strongly self-gravitating with minimal collisions.
- [Planetary rings](#) are thin discs of solid bodies where frequent collisions occur.
- [Be stars](#) are rapidly rotating high mass stars.

1 Classical Theory of Accretion Discs

1.1 Orbital dynamics

Key Points:

- Keplerian orbits with the effective potential $\Phi^{\text{eff}} = \Phi + \frac{\tilde{h}^2}{2r^2}$.
- In a Keplerian orbit,

$$h = \sqrt{GM}r, \quad \varepsilon = -\frac{GM}{2r}, \quad \Omega = \sqrt{\frac{GM}{r^3}}$$

Consider a test particle in a potential Φ using Newtonian dynamics. Let (r, ϕ, z) be cylindrical polar coordinates and assume the potential is:

1. Axisymmetric $\Phi = \Phi(r, z)$.
2. Symmetric $\Phi(r, -z) = \Phi(r, z)$.
3. Convex $\Phi_{,zz}(r, 0) > 0$.

where the subscript comma denotes partial differentiation. Assuming that the potential is that of a point mass M , we have that the potential is given by,

$$\Phi = -\frac{GM}{(r^2 + z^2)^{1/2}}$$

and orbits in this potential are described as [Keplerian orbits](#) with an equation of motion,

$$\ddot{\mathbf{r}} = -\nabla\Phi$$

We have two constraints of motion for these orbits. The first is the [specific energy](#),

$$\tilde{\varepsilon} = \frac{1}{2}|\dot{\mathbf{r}}|^2 + \Phi$$

and the second is the [specific angular momentum](#)

$$\tilde{h} = r^2\dot{\phi}$$

Using the conservation of the specific angular momentum, we can write express the equation of motion as a set of two equations,

$$\ddot{r} = -\Phi_{,r}^{\text{eff}} \quad \text{and} \quad \ddot{z} = -\Phi_{,z}^{\text{eff}}$$

where we have defined the [effective potential](#) Φ^{eff} as,

$$\Phi^{\text{eff}} = \Phi + \frac{\tilde{h}^2}{2r^2}$$

The specific energy is then found using that in cylindrical coordinates $|\dot{\mathbf{r}}|^2 = \dot{r}^2 + r^2\dot{\phi}^2 + \dot{z}^2$ combined with the specific angular momentum,

$$\tilde{\varepsilon} = \frac{1}{2}(\dot{r}^2 + r^2\dot{\phi}^2 + \dot{z}^2) + \Phi = \frac{1}{2}(\dot{r}^2 + \dot{z}^2) + \Phi^{\text{eff}}$$

Suppose the test particle dissipates energy (radiation for instance) but the angular momentum is conserved. The orbit of minimal energy for a given specific angular momentum is a circular orbit in the plane $z = 0$ at a radius $r = r_{\text{circ}}$ where $\Phi_{,r}^{\text{eff}}(r_{\text{circ}}, 0) = 0$, that is,

$$\Phi_{,r}(r_{\text{circ}}, 0) - \frac{h^2}{r_{\text{circ}}^3} = 0 \quad \text{and} \quad \varepsilon = \Phi(r_{\text{circ}}, 0) + \frac{h^2}{2r_{\text{circ}}^2}$$

with $\varepsilon(r_{\text{circ}})$ and $h(r_{\text{circ}})$ being the energy and angular momentum of the circular orbit at radius r_{circ} .

Note that for a general radius r ,

$$\begin{aligned} \frac{d\varepsilon}{dr} &= \Phi_{,r}(r, 0) - \frac{h^2}{r^3} + \frac{h}{r^2} \frac{dh}{dr} = \frac{h^2}{r^3} - \frac{h^2}{r^3} + \frac{h}{r^2} \frac{dh}{dr} = \frac{h}{r^2} \frac{dh}{dr} \\ \Rightarrow \frac{d\varepsilon}{dh} &= \frac{h}{r^2} = \dot{\phi} = \Omega \end{aligned}$$

is the angular velocity.

For Keplerian orbits in a mass potential,

$$\Phi^{\text{eff}}(r, 0) = -\frac{GM}{r} + \frac{\tilde{h}}{2r^2}$$

which gives us that,

$$h = \sqrt{GM}r, \quad \varepsilon = -\frac{GM}{2r}, \quad \Omega = \sqrt{\frac{GM}{r^3}} \quad (1)$$

1.2 Oscillations about minimal-energy orbits

Key Points:

- Perturbations of Keplerian orbits are also Keplerian orbits.

For a minimal energy orbit, $\nabla\Phi^{\text{eff}} = \mathbf{0}$. Consider perturbations $(\delta r, \delta z)$ about such an orbit at a fixed angular momentum.

Due to these perturbations, we have the following transformations,

$$\begin{aligned} r &\mapsto r' = r + \delta r \\ z &\mapsto z' = z + \delta z \end{aligned}$$

These transformations also change the effective potential as

$$(\Phi^{\text{eff}})'(r, z) = \Phi^{\text{eff}}(r', z') = \Phi^{\text{eff}}(r + \delta r, z + \delta z)$$

which we can expand to leading order in δr and δz where we assume Φ^{eff} is suitably well behaved,

$$\Phi^{\text{eff}}(r + \delta r, z + \delta z) = \Phi^{\text{eff}}(r, z) + \Phi_{,r}^{\text{eff}}(r, z)\delta r + \Phi_{,z}^{\text{eff}}(r, z)\delta z + \mathcal{O}(\delta r, \delta z)$$

Consequently, the first of our equations of motion now reads as,

$$\ddot{r}' = -(\Phi^{\text{eff}})'_{,r} \Rightarrow \ddot{r} + \delta\ddot{r} = -\Phi_{,r}^{\text{eff}} - \Phi_{,rr}^{\text{eff}}\delta r \Rightarrow \delta\ddot{r} = -\Phi_{,rr}^{\text{eff}}\delta r$$

Using the equation of motion for r and since we are in a circular orbit (minimal energy orbit), we further have that,

$$\delta\ddot{r} = -\kappa^2\delta r$$

where we have defined the [epicyclic frequency](#) $\kappa(r)$ as,

$$\kappa^2(r) := \Phi_{,rr}^{\text{eff}}(r, 0)$$

Similarly, the second of our equations of motion now reads as,

$$\ddot{z}' = -(\Phi^{\text{eff}})'_{,z} \Rightarrow \ddot{z} + \delta\ddot{z} = -\Phi_{,z}^{\text{eff}} - \Phi_{,zz}^{\text{eff}}\delta z \Rightarrow \delta\ddot{z} = -\Phi_{,zz}^{\text{eff}}\delta z$$

and we further have that,

$$\delta\ddot{z} = -\Omega_z^2\delta z$$

where we have defined the **vertical frequency** $\Omega_z(r)$ as,

$$\Omega_z^2(z) := \Phi_{,zz}^{\text{eff}}(r, 0)$$

Moreover, using our definition for the effective potential,

$$\kappa^2 = \Phi_{,rr} + \frac{3h^2}{r^4} = \frac{d}{dr} \left(\frac{h^2}{r^3} \right) + \frac{3h^2}{r^4} = \frac{1}{r^3} \frac{dh^2}{dr}$$

which is more often written as,

$$\kappa^2 = 4\Omega^2 + 2r\Omega \frac{d\Omega}{dr}$$

Similarly,

$$\Omega_z^2 = \Phi_{,zz}(r, 0)$$

For a point mass potential, $\kappa = \Omega_z = \Omega$ and these are known as **inclined Keplerian orbits**.

1.3 Angular momentum redistribution and energy dissipation

Key Points:

- For fixed mass, energy is released by transferring angular momentum to the particle with smaller angular velocity.
- For variable mass, energy is released by transferring angular momentum outwards and mass inwards.

Consider two particles m_1 and m_2 with specific angular momenta h_1 and h_2 .

Our total energy is therefore,

$$E = E_1 + E_2 = m_1\varepsilon_1 + m_2\varepsilon_2$$

and our total angular momentum is,

$$H = H_1 + H_2 = m_1h_1 + m_2h_2$$

Taking the differential of our total energy,

$$dE = dE_1 + dE_2 = m_1\Omega_1 dh_1 + m_2\Omega_2 dh_2$$

where we have used that $d\varepsilon = \Omega dh$. Similarly, taking the differential of our total angular momentum,

$$dH = dH_1 + dH_2 = m_1 dh_1 + m_2 dh_2$$

The conservation of angular momentum gives the constraint $dH = 0$. Using this, we can express the differential of the total energy in terms of (say) the differential of particle m_1 's angular momentum,

$$dH = 0 \Rightarrow dH_2 = -dH_1 \Rightarrow dE = (\Omega_1 - \Omega_2) dH_1$$

Thus energy is released by transferring angular momentum to the particle with smaller angular velocity Ω ; this is since $\Omega_1 - \Omega_2$ has the opposite sign to dH_1 when angular momentum is transferred from the faster-rotating particle to the slower-rotating particle, giving $dE < 0$.

Generalise this argument to allow mass to be exchanged. The differential of the total mass and angular momentum is given by,

$$dM = dm_1 + dm_2 = 0$$

$$dH = dH_1 + dH_2 = 0$$

which are both conserved, with $dH_i = m_i dh_i + h_i dm_i$ courtesy of the product rule for $i = 1, 2$. Now applying the same argument, we can express the differential of the energy of particle m_1 using the product rule as,

$$\begin{aligned} dE_1 &= m_1 d\varepsilon_1 + \varepsilon_1 dm_1 = m_1 \Omega_1 dh_1 + \varepsilon_1 dm_1 \\ &= \Omega_1 (dH_1 - h_1 dm_1) + \varepsilon_1 dm_1 \\ &= \Omega_1 dH_1 + (\varepsilon_1 - \Omega_1 h_1) dm_1 \end{aligned}$$

where we have used that $m_1 dh_1 = dH_1 - h_1 dm_1$. We can now write the differential of the total energy of the particles as,

$$\begin{aligned} dE &= \Omega_1 dH_1 + (\varepsilon_1 - \Omega_1 h_1) dm_1 + \Omega_2 dH_2 + (\varepsilon_2 - \Omega_2 h_2) dm_2 \\ &= (\Omega_1 - \Omega_2) dH_1 + [(\varepsilon_1 - \Omega_1 h_1) - (\varepsilon_2 - \Omega_2 h_2)] dm_1 \end{aligned}$$

where we used the same argument as the previous for dE_2 , $dm_2 = -dm_1$ and also that $dH_2 = -dH_1$. Finally notice that,

$$\frac{d}{dr}(\varepsilon - \Omega h) = \frac{h}{r^2} \frac{dh}{dr} - h \frac{d\Omega}{dr} - \Omega \frac{dh}{dr} = -h \frac{d\Omega}{dr}$$

In most astrophysical rotating systems we're considering, the angular velocity decreases as you move further from the centre, that is $\frac{d\Omega}{dr} < 0$. Thus $\frac{d}{dr}(\varepsilon - \Omega h) > 0$ and energy is released by transferring angular momentum outwards and mass inwards.

1.4 Viscous evolution of an accretion disc

Key Points:

- The 1 dimensional conservation of mass equation is

$$\frac{\partial \Sigma}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r \Sigma \bar{u}_r) = 0$$

where $\Sigma(r, t)$ is the surface density.

- The 1 dimensional conservation of angular momentum equation is

$$\Sigma \bar{u}_r \frac{dh}{dr} = \frac{1}{r} \frac{\partial}{\partial r} \left(\bar{\nu} \Sigma r^2 \frac{d\Omega}{dr} \right).$$

- The diffusion equation for the surface density is

$$\frac{\partial \Sigma}{\partial t} = \frac{3}{r} \frac{\partial}{\partial r} \left[r^{1/2} \frac{\partial}{\partial r} (r^{1/2} \bar{\nu} \Sigma) \right].$$

- If $\bar{\nu} = \bar{\nu}(r)$, then the above diffusion equation can be written as

$$\frac{\partial \mathcal{G}}{\partial t} = \bar{\nu} r^2 \frac{dh}{dt} \left(-\frac{d\Omega}{dr} \right) \frac{\partial^2 \mathcal{G}}{\partial h^2}$$

where $\mathcal{G}(r, t)$ is the viscous torque.

1.4.1 Introduction

The evolution of an accretion disc is regulated by two conservation laws: conservation of mass and conservation of angular momentum. Our aim is to reduce these 3 dimensional laws into 1 dimensional equations.

From fluid dynamics, the equation of mass conservation is

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

where ρ is the density of the fluid and $\mathbf{u}$ is the velocity of the fluid.

The equation of motion is,

$$\rho \frac{D\mathbf{u}}{Dt} = -\rho \nabla \Phi - \nabla p + \nabla \cdot \mathbf{T}$$

where $\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla$ is the [Lagrangian time-derivative](#), Φ is the gravitational potential, p is the pressure and $\mathbf{T}$ is the [stress tensor](#).

We will employ cylindrical coordinates (r, ϕ, z) for all our calculations with the central mass is located at $r = z = 0$ and the mid-plane of the disc is at $z = 0$.

The stress tensor $\mathbf{T}$ could be turbulent stress due to tangled magnetic fields or other contributions. In the classical theory, the stress is parametrised by the viscous stress.

1.4.2 Conservation of mass

Define the [surface density](#) $\Sigma(r, t)$ by,

$$\Sigma(r, t) = \frac{1}{2\pi} \int_0^{2\pi} \int_{-\infty}^{\infty} \rho \, dz \, d\phi$$

The mass contained between r_1 and r_2 is given by,

$$\int_V \rho \, dV = \int_{r_1}^{r_2} \int_0^{2\pi} \int_{-\infty}^{\infty} r \rho \, dz \, d\phi \, dr = \int_{r_1}^{r_2} 2\pi r \Sigma \, dr$$

In cylindrical coordinates, the equation of mass conservation reads,

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r \rho u_r) + \frac{1}{r} \frac{\partial}{\partial \phi} (\rho u_\phi) + \frac{\partial}{\partial z} (\rho u_z) = 0$$

which we can multiply by r and integrate over ϕ, z over the full disc,

$$\begin{aligned} r \frac{\partial}{\partial t} \left[\int_0^{2\pi} \int_{-\infty}^{\infty} \rho \, dz \, d\phi \right] + \frac{\partial}{\partial r} \left[\int_0^{2\pi} \int_{-\infty}^{\infty} r \rho u_r \, dz \, d\phi \right] \\ + \int_0^{2\pi} \int_{-\infty}^{\infty} \frac{\partial}{\partial \phi} (\rho u_\phi) \, dz \, d\phi + r \int_0^{2\pi} \int_{-\infty}^{\infty} \frac{\partial}{\partial z} (\rho u_z) \, dz \, d\phi = 0 \end{aligned}$$

Defining the [radial mass flux](#) $\mathcal{F}(r, t)$ by,

$$\mathcal{F}(r, t) = \int_0^{2\pi} \int_{-\infty}^{\infty} r \rho u_r \, dz \, d\phi$$

and under two assumptions: (1) axial-symmetry ($\frac{\partial}{\partial \phi} = 0$) and (2) $\rho u_z \rightarrow 0$ as $z \rightarrow \pm\infty$; the latter assumption being the assumption that there is no vertical loss of mass to $z = \pm\infty$, we find that,

$$2\pi \frac{\partial \Sigma}{\partial r} + \frac{1}{r} \frac{\partial \mathcal{F}}{\partial r} = 0$$

In addition, we can also define a density-weighted [mean radial velocity](#) $\bar{u}_r(r, t)$ by,

$$\mathcal{F} = 2\pi r \Sigma \bar{u}_r$$

Using all of the above, we have reduced the 3 dimensional mass conservation to the 1 dimensional equation,

$$\frac{\partial \Sigma}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r \Sigma \bar{u}_r) = 0 \tag{2}$$

1.4.3 Conservation of angular momentum

The azimuthal component of the conservation of momentum reads,

$$\rho \left(\frac{Du_\phi}{Dt} + \frac{u_r u_\phi}{r} \right) = -\frac{\rho}{r} \frac{\partial \Phi}{\partial \phi} - \frac{1}{r} \frac{\partial p}{\partial \phi} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 T_{r\phi}) + \frac{1}{r} \frac{\partial T_{\phi\phi}}{\partial \phi} + \frac{\partial T_{\phi z}}{\partial z}$$

and, under an axisymmetric potential $\Phi = \Phi(r, z)$, this becomes

$$\rho \frac{D}{Dt} (ru_\phi) = \frac{1}{r} \frac{\partial}{\partial \phi} (-rp + rT_{\phi\phi}) + \frac{1}{r} \frac{\partial}{\partial r} (r^2 T_{r\phi}) + \frac{\partial}{\partial z} (rT_{\phi z})$$

where we have used

$$\begin{aligned} \rho \frac{D}{Dt} (ru_\phi) &= r\rho \frac{Du_\phi}{Dt} + u_\phi \rho \frac{Dr}{Dt} \\ &= r\rho \frac{Du_\phi}{Dt} + u_\phi \rho \left[\frac{\partial r}{\partial t} + \mathbf{u} \cdot \nabla r \right] \\ &= r\rho \frac{Du_\phi}{Dt} + u_\phi \rho \left[(u_r \mathbf{e}_r + u_\phi \mathbf{e}_\phi + u_z \mathbf{e}_z) \cdot \left(\frac{\partial r}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial r}{\partial \phi} \mathbf{e}_\phi + \frac{\partial r}{\partial z} \mathbf{e}_z \right) \right] \\ &= r\rho \frac{Du_\phi}{Dt} + \rho u_\phi u_r \end{aligned}$$

For an axisymmetric disc, $\frac{\partial}{\partial \phi} = 0$, and under the thin-disc approximation, the vertical transport of angular momentum is negligible compared with the radial transport. Therefore,

$$\rho \frac{D}{Dt} (ru_\phi) = \frac{1}{r} \frac{\partial}{\partial r} (r^2 T_{r\phi})$$

Consequently, we have that angular momentum is conserved and the term $T_{r\phi}$ describes the radial transport of angular momentum through viscous stresses.

Now assume that $u_\phi = r\Omega$, where $\Omega(r)$ is the angular velocity of circular orbits in the potential Φ ; this leads to,

$$\rho \frac{D}{Dt} (r^2 \Omega) = \frac{1}{r} \frac{\partial}{\partial \phi} (-rp + rT_{\phi\phi}) + \frac{1}{r} \frac{\partial}{\partial r} (r^2 T_{r\phi}) + \frac{\partial}{\partial z} (rT_{\phi z})$$

Since $h = r^2 \Omega$ is the specific orbital angular momentum,

$$\frac{D(r^2 \Omega)}{Dt} = \frac{Dh}{Dt} = \frac{\partial h}{\partial t} + \mathbf{u} \cdot \nabla h = u_r \frac{dh}{dr}$$

and hence,

$$\rho u_r \frac{dh}{dr} = \frac{1}{r} \frac{\partial}{\partial \phi} (-rp + rT_{\phi\phi}) + \frac{1}{r} \frac{\partial}{\partial r} (r^2 T_{r\phi}) + \frac{\partial}{\partial z} (rT_{\phi z})$$

Multiplying by r and integrating with respect to ϕ and z ,

$$\begin{aligned} & \int_0^{2\pi} \int_{-\infty}^{\infty} r \rho u_r \frac{dh}{dr} dz d\phi \\ &= \int_0^{2\pi} \int_{-\infty}^{\infty} \left[\frac{\partial}{\partial \phi} (-rp + rT_{\phi\phi}) + \frac{\partial}{\partial r} (r^2 T_{r\phi}) + \frac{\partial}{\partial z} (r^2 T_{\phi z}) \right] dz d\phi \end{aligned}$$

Defining the **viscous torque** $\mathcal{G}(r, t)$ as,

$$\mathcal{G} = - \int_0^{2\pi} \int_{-\infty}^{\infty} r^2 T_{r\phi} dz d\phi$$

we obtain

$$\mathcal{F} \frac{dh}{dr} = \int_{-\infty}^{\infty} [-rp + rT_{\phi\phi}]_0^{2\pi} dz + \int_0^{2\pi} [r^2 T_{\phi z}]_{-\infty}^{\infty} d\phi + \frac{\partial \mathcal{G}}{\partial r}$$

The first term vanishes because all physical quantities are periodic in ϕ , while the second vanishes from the assumption that there is no vertical loss of angular momentum as $z \rightarrow \pm\infty$. Therefore, we yield the relation,

$$\mathcal{F} \frac{dh}{dr} = - \frac{\partial \mathcal{G}}{\partial r}$$

In an astrophysical discs, the molecular viscosity is generally too small to account for the observed transport of angular momentum. We therefore introduce an effective viscosity, which parametrises the processes responsible for this transport. The viscous force acting within the fluid is described by the viscous stress tensor $\mathbf{T}$.

For a Newtonian fluid, the Navier–Stokes (NS) constitutive relation gives

$$\mathbf{T} = \mu [\nabla \mathbf{u} + (\nabla \mathbf{u})^T] + \left(\mu_b - \frac{2}{3} \mu \right) (\nabla \cdot \mathbf{u}) \mathbf{1}$$

where μ is the **dynamic viscosity**, μ_b is the **bulk viscosity** and $\mathbf{1}$ is the identity tensor. The dynamic viscosity is related to the kinematic viscosity ν by $\mu = \rho\nu$.

For a thin disc in circular, axisymmetric motion, $\mathbf{u} = (0, r\Omega, 0)$ in the cylindrical coordinate basis and $\frac{\partial}{\partial \phi} = 0$. The relevant stress for the transport of angular momentum is the $r\phi$ component,

$$T_{r\phi} = \mu \left(\frac{\partial u_\phi}{\partial r} - \frac{u_\phi}{r} + \frac{1}{r} \frac{\partial u_r}{\partial \phi} \right)$$

Under the assumptions above, this reduces to

$$T_{r\phi} = \mu \left[\frac{\partial}{\partial r} (r\Omega) - \Omega \right] = \mu r \frac{d\Omega}{dr}$$

Since the viscous stress tensor is symmetric, the only non-zero components of the viscous stress tensor $\mathbf{T}$ is,

$$T_{r\phi} = T_{\phi r} = \mu r \frac{d\Omega}{dr}$$

Define the density-weighted [mean kinematic viscosity](#) $\bar{\nu}(r, t)$ by,

$$\bar{\nu}\Sigma = \frac{1}{2\pi} \int_0^{2\pi} \int_{-\infty}^{\infty} \mu \, dz \, d\phi$$

which gives our viscous torque as,

$$\mathcal{G} = -2\pi\bar{\nu}\Sigma r^3 \frac{d\Omega}{dr}$$

Although this is derived from the NS viscosity, the torque can always be parametrised in this form for some $\bar{\nu}(r, t)$.

Using all of the above, we have reduced the 3 dimensional (angular) momentum conservation into the 1 dimensional equation,

$$\Sigma \bar{u}_r \frac{dh}{dr} = \frac{1}{r} \frac{\partial}{\partial r} \left(\bar{\nu} \Sigma r^3 \frac{d\Omega}{dr} \right) \quad (3)$$

1.4.4 Diffusion equation for surface density

Equations (2) and (3) can be combined to eliminate $\bar{u}_r$,

$$\frac{\partial \Sigma}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} \left[\left(\frac{dh}{dr} \right)^{-1} \frac{\partial}{\partial r} \left(\bar{\nu} \Sigma r^3 \frac{d\Omega}{dr} \right) \right] = 0$$

For a point-mass potential, we have $\Omega = \left(\frac{GM}{r^3} \right)^{1/2}$ and $h = (GMr)^{1/2}$ from subsection 1.1. As a result,

$$\frac{\partial \Sigma}{\partial t} = \frac{3}{r} \frac{\partial}{\partial r} \left[r^{1/2} \frac{\partial}{\partial r} (r^{1/2} \bar{\nu} \Sigma) \right] \quad (4)$$

which is the [diffusion equation for the surface density](#) $\Sigma(r, t)$ and can be interpreted as follows. Viscous torques cause the redistribution of angular momentum, which causes a viscous ‘spreading’ or diffusion. Most of the mass moves to smaller radii, but some of the mass moves to larger radii.

If the mean kinematic viscosity is independent of time, $\bar{\nu} = \bar{\nu}(r)$, then the diffusion equation for the surface density reads,

$$\frac{\partial \mathcal{G}}{\partial t} = \bar{\nu} r^2 \frac{dh}{dt} \left(-\frac{d\Omega}{dr} \right) \frac{\partial^2 \mathcal{G}}{\partial h^2}$$

which is an alternate form describing the torque diffusing in the space of specific angular momentum.

1.5 Analysis of the diffusion equation

1.5.1 Inner boundary conditions

The inner boundary conditions depend on the nature of the object, we will discuss the following: (1) weakly-magnetized stars (2) black holes (3) strongly-magnetized stars.

Weakly-magnetized stars

For a non(weakly)-magnetized star, the accretion disc may extend to the stellar surface. Usually the star rotates at only a fraction of the Keplerian angular velocity at its surface, that is,

$$\Omega_\star < \left(\frac{GM}{R_\star^3} \right)^{1/2}$$

due to the star mainly being supported by pressure and not the centrifugal force. The angular velocity of the fluid in the accretion disc makes a rapid transition from Ω to $\Omega_\star$ in a [viscous boundary layer](#). In this boundary layer, there exists a radius r_{in} where the shear rate is zero and $\mathcal{G} = 0$. This radius is regarded as the inner radius of the disc and $r_{\text{in}} \approx R_\star$.

Black holes

For a black hole, the accretion disc does not extend to the horizon due to the existence of a marginally stable circular orbit $r = r_{\text{ms}}$. For $r < r_{\text{ms}}$, circular orbits are unstable and spiral into the black hole.

For a non-rotating (Schwarzschild) black hole, $r_{\text{ms}} = 6GM/c^2$ and the event horizon is given by $r_S = 2GM/c^2$. In order to conserve mass, the surface density Σ decreases rapidly just inside r_{ms} , and the viscous stress vanishes at r_{ms} .

Strongly-magnetized stars

For strongly-magnetized stars, the inner part of the disc is in fact disrupted by the magnetic field. This causes the disc to terminate at the so called [magneto-spherical radius](#).

Summary

The usual inner boundary conditions are $\mathcal{G} = 0$ at $r = r_{\text{in}}$. If the angular velocity is Keplerian throughout the disc, then this implies,

$$r^{1/2} \bar{\nu} \Sigma = 0 \quad \text{at} \quad r = r_{\text{in}}$$

1.5.2 Steady Solutions

If the disc is in a steady state, that is $\frac{\partial}{\partial t} = 0$, then the conservation of mass equation reads,

$$\frac{1}{r} \frac{\partial}{\partial r} (r \Sigma \bar{u}_r) = 0$$

which we can write in terms of the radial mass flux $\mathcal{F}$,

$$\mathcal{F} = \text{constant} = -\dot{M}$$

where we have defined the **mass accretion rate** $\dot{M}$. The conservation of angular momentum equation reads,

$$\mathcal{F} \frac{dh}{dr} = -\frac{\partial \mathcal{G}}{\partial r} \Rightarrow -\dot{M}h + \mathcal{G} = \text{constant}$$

and we can find this constant using the boundary conditions, $\mathcal{G} = 0$ at $r = r_{\text{in}}$, to yield,

$$\mathcal{G} = \dot{M}(h - h_{\text{in}})$$

where $h_{\text{in}} = h(r_{\text{in}})$.

Recall that, for Keplerian discs we have $h = (GM/r)^{1/2}$ and $\Omega = (GM/r^3)^{1/2}$ and our expression for the viscous torque in terms of the mean kinematic viscosity,

$$\mathcal{G} = -2\pi\bar{\nu}\Sigma r^3 \frac{d\Omega}{dr}$$

using this and the previous expression for $\mathcal{G}$, we find that

$$\bar{\nu}\Sigma = \frac{\dot{M}}{3\pi} \left[1 - \left(\frac{r_{\text{in}}}{r} \right)^{1/2} \right]$$

which if $\bar{\nu}(r, \Sigma)$ is known, then this gives a complete solution for the disc.

1.5.3 Spectrum of emitted radiation

This section particularly has a lot of assumed knowledge, mainly of statistical physics.

Taking as a given from fluid mechanics under our calculations in subsection 1.4.3, the rate of energy dissipation per unit volume is given by,

$$T_{r\phi} r \frac{d\Omega}{dr} = \mu \left(r \frac{d\Omega}{dr} \right)^2$$

Assuming the this is radiated away locally in radius, instead of being advected through the disc or retained in the gas, then the rate at which energy is emitted (energy flux) per unit area of each face of the disc is *half* of the integral over z and ϕ with the normalising factor of $1/(2\pi)$,

$$\frac{1}{2} \left(\frac{1}{2\pi} \int_0^{2\pi} \int_{-\infty}^{\infty} \mu \left(r \frac{d\Omega}{dr} \right)^2 dz d\phi \right) = \frac{1}{2} \bar{\nu}\Sigma \left(r \frac{d\Omega}{dr} \right)^2$$

If this is carried by black-body radiation¹ in a Keplerian disc, then from the Stefan-Boltzmann law we know that the energy flux is related to the surface temperature $T_{\text{eff}}(r, t)$ as,

$$\frac{1}{2}\bar{\nu}\Sigma\left(r\frac{d\Omega}{dr}\right)^2 \stackrel{\text{Kep}}{=} \frac{9}{8}\bar{\nu}\Sigma\Omega^2 = \sigma T_{\text{eff}}^4(r, t)$$

where σ is the Stefan-Boltzmann constant. If the disc is not black-bodied, then this can be taken as the definition of the **effective temperature** of the disc.

Using our previous expression for $\bar{\nu}\Sigma$ in a steady Keplerian disc, we have that,

$$T_{\text{eff}}^4(r) = \frac{3GM\dot{M}}{8\pi\sigma r^3} \left[1 - \left(\frac{r_{\text{in}}}{r}\right)^{1/2}\right]$$

If the radiation has a black-body spectrum², then by definition each annulus has a specific emissivity proportional to the Planck function,

$$B_{\nu}[T_{\text{eff}}] \propto \frac{\nu^3}{\exp(h\nu/kT_{\text{eff}}) - 1}$$

where ν here is the frequency, h is Planck's constant and k is Boltzmann's constant. The total spectrum is obtained by integrating over all annuli,

$$F_{\nu} \propto \int_{r_{\text{in}}}^{r_{\text{out}}} B_{\nu}[T_{\text{eff}}(r)] \cdot 2\pi r \, dr$$

where the $2\pi r \, dr$ comes from the area of an annulus dA . This gives a stretched spectrum, with the low-frequency emission comes mainly from the outer part of the disc, and the high-frequency emission from the inner part.

The total luminosity of a disc stretching to $r_{\text{out}} = \infty$ is given by,

$$L_{\text{disc}} = \int_{r_{\text{in}}}^{\infty} \underbrace{\sigma T_{\text{eff}}^4}_{\sigma T^4} \cdot \underbrace{2\pi r \, dr}_{dA} = \frac{1}{2} \frac{GM\dot{M}}{r_{\text{in}}}$$

Which is exactly equal to the rate at which orbital binding energy is transferred to the gas as it passes from infinity to r_{in} ³. But it is only half the potential energy released. In the case of accretion on to a stellar surface, the remaining energy is released in the boundary layer.

1.5.4 Time dependent solutions

For a Keplerian disc, the diffusion equation reads,

$$\frac{\partial \Sigma}{\partial t} = \frac{3}{r} \frac{\partial}{\partial r} \left[r^{1/2} \frac{\partial}{\partial r} (r^{1/2} \bar{\nu} \Sigma) \right]$$

with the constraint $r^{1/2} \bar{\nu} \Sigma = 0$ at $r = r_{\text{in}}$. Suppose $\bar{\nu}(r, \Sigma)$ is given, we have two options:

¹That is a body that absorbs all electromagnetic radiation.

²The distribution of electromagnetic radiation emitted by a black body

³Recall binding energy from subsection 0.1

1. $\bar{\nu} = \bar{\nu}(r)$ and we have a linear diffusion equation.
2. $\bar{\nu} = \bar{\nu}(r, \Sigma)$ and we have a non-linear diffusion equation.

For simplicity, we will assume $r_{\text{in}} \rightarrow 0$ which gives the constraints as $r^{1/2}\bar{\nu}\Sigma \rightarrow 0$ as $r \rightarrow 0$.

Linear case

Suppose that we have $\bar{\nu} = \bar{\nu}(r)$. Using separation of variables, assume a solution of the form,

$$\Sigma(r, t) = R(r)T(t)$$

Substituting this into our diffusion equation (4) and simplifying,

$$\frac{\dot{T}(t)}{T(t)} = \frac{3}{rR(r)} \left[r\bar{\nu}R''(r) + \left(\frac{3}{2}\bar{\nu} + 2r\bar{\nu}' \right) R'(r) + \left(\frac{3}{2}\bar{\nu}' + r\bar{\nu}'' \right) R(r) \right]$$

which leads to each side of the equation being equal to the same constant λ . Solving the temporal mode gives,

$$T(t) \propto e^{-\lambda t}$$

while the spacial mode is slightly harder. We can make two further assumptions that will make our lives easier, the first is that we will assume $\bar{\nu} = \text{constant}$ and the second being $R(r) = \sigma(r)r^\beta$ where β is a free parameter. The first assumption seems fair, the second may seem out of nowhere. The justification may be thought as follows: the witting of $R(r) = \sigma(r)r^\beta$ can be interpreted as ‘factoring’ a power of r from the series expansion of $R(r)$, leaving us with a function $\sigma(r)$.

Using these two further assumptions and $\beta = -1/4$, we can write,

$$r^2 \frac{d^2\sigma}{dr^2} + r \frac{d\sigma}{dr} + \left(k^2 r^2 - \frac{1}{16} \right) \sigma = 0$$

where $k^2 = \lambda/(3\bar{\nu})$, which is *Bessel’s equation* of order $1/4$. It is known that this equation has a general solution,

$$\sigma(r) = A\mathcal{J}_{1/4}(kr) + B\mathcal{Y}_{1/4}(kr)$$

where $\mathcal{J}_{1/4}(kr)$ and $\mathcal{Y}_{1/4}(kr)$ are the Bessel functions of the first and second kind respectively. Moreover, it is known that for small r ,

$$\sigma(r) \propto Ar^{1/4} + Br^{-1/4}$$

from the properties of the Bessel functions. This gives the following expression,

$$r^{1/2}\bar{\nu}\Sigma \propto (Ar^{1/2} + B)e^{-\lambda t} \xrightarrow{r \rightarrow 0} Be^{-\lambda t}$$

which, under our initial conditions, we require to vanish, and so $B = 0$. Finally, we obtain the surface density as,

$$\Sigma(r, t) \propto r^{-1/4} \mathcal{J}_{1/4}(kr) e^{-3\bar{\nu}k^2t}$$

for the linear case where $\bar{\nu} = \text{constant}$.

Now, consider a more general initial-value problem. Since the functions $r^{-1/4} \mathcal{J}_{1/4}(kr)$ form a complete orthogonal set, the initial surface-density profile can be expanded in this basis as

$$\Sigma(r, 0) = \int_0^\infty f(k) r^{-1/4} \mathcal{J}_{1/4}(kr) dk,$$

where $f(k)$ is the corresponding expansion coefficient, determined by the initial condition.

The general solution is then obtained by evolving each mode independently, as above. Since the governing equation is linear,

$$\Sigma(r, t) = \int_0^\infty f(k) r^{-1/4} \mathcal{J}_{1/4}(kr) T(k, t) dk$$

where $T(k, t)$ describes the time evolution of an individual mode. Thus, the solution is constructed as a superposition of Bessel modes, each evolving independently according to the solution derived above. From previous, we know the form of $T(k, t)$,

$$\Sigma(r, t) = \int_0^\infty f(k) r^{-1/4} \mathcal{J}_{1/4}(kr) e^{-3\bar{\nu}k^2t} dk$$

We can find the form of $f(k)$ in a clever way. Given the Bessel functions of order ν , we can define the *Hankel transformations* as the pair,

$$\begin{aligned} A(r) &= \int_0^\infty a(k) \mathcal{J}_\nu(kr) (kr)^{1/2} dk \\ a(k) &= \int_0^\infty A(r) \mathcal{J}_\nu(kr) (kr)^{1/2} dr \end{aligned}$$

for some ‘nice’ $A(r)$ and $a(k)$. We can strategically write,

$$r^{3/4} \Sigma(r, 0) = \int_0^\infty k^{-1/2} f(k) \mathcal{J}_{1/4}(kr) (kr)^{1/2} dk$$

and it follows from our Hankel transformations that,

$$k^{-1/2} f(k) = \int_0^\infty s^{3/4} \Sigma(s, 0) \mathcal{J}_{1/4}(ks) (ks)^{1/2} ds$$

which we can use to write,

$$\begin{aligned}
\Sigma(r, t) &= \int_0^\infty \left[k^{1/2} \int_0^\infty s^{3/4} \Sigma(s, 0) \mathcal{J}_{1/4}(ks) (ks)^{1/2} ds \right] r^{-1/4} \mathcal{J}_{1/4}(kr) e^{-3\bar{\nu}k^2t} dk \\
&= \int_0^\infty \int_0^\infty k^{1/2} s^{3/4} \Sigma(s, 0) \mathcal{J}_{1/4}(ks) (ks)^{1/2} r^{-1/4} \mathcal{J}_{1/4}(kr) e^{-3\bar{\nu}k^2t} ds dk \\
&= \int_0^\infty \left(\int_0^\infty k^{1/2} s^{3/4} \mathcal{J}_{1/4}(ks) (ks)^{1/2} r^{-1/4} \mathcal{J}_{1/4}(kr) e^{-3\bar{\nu}k^2t} dk \right) \Sigma(s, 0) ds \\
&= \int_0^\infty G(r, s, t) \Sigma(s, 0) ds
\end{aligned}$$

where we swapped the order of integration and defined the *Green's function* $G(r, s, t)$ as,

$$\begin{aligned}
G(r, s, t) &= \int_0^\infty k^{1/2} s^{3/4} \mathcal{J}_{1/4}(ks) (ks)^{1/2} r^{-1/4} \mathcal{J}_{1/4}(kr) e^{-3\bar{\nu}k^2t} dk \\
&= r^{-1/4} s^{5/4} \frac{1}{6\bar{\nu}t} \exp\left(-\frac{(r^2 + s^2)}{12\bar{\nu}t}\right) \mathcal{I}_{1/4}\left(\frac{rs}{6\bar{\nu}t}\right)
\end{aligned}$$

where $\mathcal{I}_{1/4}(\cdot)$ is the *modified Bessel function of the first order*.

Eventually, all the mass ends up on the central object and the angular momentum is carried out to infinity by a negligible quantity of gas.

Non-linear case

Consider the following transformation of the diffusion equation. Let $x = r^{1/2}$, using the chain rule we obtain,

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r} \frac{\partial}{\partial x} = \frac{1}{2x} \frac{\partial}{\partial x}$$

which gives (4) in terms of x ,

$$\frac{\partial \Sigma}{\partial t} = \frac{3}{4x^3} \frac{\partial^2}{\partial x^2} (x\bar{\nu}\Sigma) \quad (5)$$

From subsection 1.4.2, the total mass of the total mass of the disc is given by,

$$\int_0^{r_{\text{out}}} 2\pi r \Sigma dr \propto \int_0^{x_{\text{out}}} \Sigma x^3 dx$$

where we have used our definition $x = r^{1/2}$ and the relation $dr = 2x dx$. We will apply a similar argument for the total angular momentum. In a Keplerian disc, the specific angular momentum is $h = (GMr)^{1/2}$, and an annulus of the disc between r and $r + dr$ has a mass

$$dM = 2\pi r \Sigma dr$$

Therefore, a disc between r and $r + dr$ has an angular momentum

$$2\pi r \Sigma (GMr)^{1/2} dr$$

and thus the total angular momentum of the disc is

$$\int_0^{r_{\text{out}}} 2\pi r \Sigma (GM r)^{1/2} dr \propto \int_0^{x_{\text{out}}} \Sigma x^4 dx$$

again using the same definition $x = r^{1/2}$ and relation $dr = 2x dx$. These quantities, the total mass and angular momentum of the disc, are in fact **locally** conserved. We will show an argument for the total mass and this argument can be replicated for the total angular momentum of the disc.

Notice that we can write (5) as

$$\frac{\partial}{\partial t}(\Sigma x^3) + \frac{\partial}{\partial x} \left[-\frac{\partial}{\partial x} \left(\frac{3}{4} x \bar{\nu} \Sigma \right) \right] = 0$$

This is enough to deduce a conservation due to its form of

$$\frac{\partial}{\partial t}(\text{Density}) + \frac{\partial}{\partial x}(\text{Flux}) = 0$$

which produces the same calculation of

$$\frac{d}{dt} \left(\int_a^b \text{Density} dx \right) = [\text{Flux}]_a^b$$

which shows it is locally conserved as the density depends only on the incoming and exiting flux through the boundary.

Our boundary conditions require $x \bar{\nu} \Sigma \rightarrow 0$ as $x \rightarrow 0$. However, this does not mean that the mass flux through the inner boundary is necessarily zero.

For instance, if we have $x \bar{\nu} \Sigma \propto x$, then the boundary conditions are satisfied. However, its derivative with respect to x does not necessarily vanish.

We are looking for a solution in which mass is accreted at the origin, but no torque is exerted there. Assuming a free boundary at x_{out} , then the only **globally** conserved quantity is

$$\int_0^{x_{\text{out}}} \Sigma x^4 dx = C$$

which is proportional to the total angular momentum of the disc.

Consider an initial-value problem for the diffusion equation (4) with $\bar{\nu} = A r^a \Sigma^b$, and A and C are the only dimensional constants in the problem. We will now look for a [similarity solution](#), that is a solution to a differential equation whose shape stays the same as time evolves, but whose scale changes.

For the similarity solution, we wish for a [similarity variable](#). For our disc setting, that will be $\xi = x/X(t)$ where $X(t)$ is built from the dimensional constants and time

$$X(t) = A^\alpha C^\beta t^\gamma$$

We will also impose $[X] = \mathbf{L}^{1/2}$ so that ξ is dimensionless.

Under the viscosity law,

$$\bar{\nu} = Ax^{15/7}\Sigma^{3/7} \quad (6)$$

we can use dimensional analysis to find that,

$$\begin{aligned} [A] &= \mathbf{M}^{-3/7} \mathbf{L}^{25/14} \mathbf{T}^{-1} \\ [C] &= \mathbf{M} \mathbf{L}^{1/2} \end{aligned}$$

where we used $[\bar{\nu}] = \mathbf{L}^2 \mathbf{T}^{-1}$, $[\Sigma] = \mathbf{M} \mathbf{L}^{-2}$ and $[x] = [\mathrm{d}x] = \mathbf{L}^{1/2}$. We can then apply dimensional consistency for $[X] = \mathbf{L}^{1/2}$, which gives,

$$X(t) = A^{1/4} C^{3/28} t^{1/4}$$

and accordingly, our similarity variables is now,

$$\xi = xA^{-1/4}C^{-3/28}t^{-1/4}$$

Since we are looking for a similarity solution and the only dimensional quantities are A , C and t , we write,

$$\bar{\nu}\Sigma = f(\xi)A^\delta C^\epsilon t^\zeta$$

for some dimensionless function $f(\xi)$, and use dimensional analysis to find,

$$\bar{\nu}\Sigma = f(\xi)A^{-1/4}C^{25/28}t^{-5/4}$$

Using the viscosity law (6), we can invert this expression for the surface density,

$$\Sigma = A^{-7/8}C^{5/8}x^{-3/2}t^{-7/8}f^{7/10}$$

and substituting this back into the diffusion equation (4) and using the chain rule, we obtain,

$$\frac{3}{4}\xi^{-4}\frac{\mathrm{d}}{\mathrm{d}\xi}\left(\xi^2\frac{\mathrm{d}f}{\mathrm{d}\xi}\right) = \xi^{-3/2}\left(-\frac{7}{8}f^{7/10} - \frac{1}{4}\xi\frac{\mathrm{d}}{\mathrm{d}\xi}f^{7/10}\right)$$

which we can write as a total derivative,

$$\frac{\mathrm{d}}{\mathrm{d}\xi}\left(3\xi^2\frac{\mathrm{d}f}{\mathrm{d}\xi} + \xi^{7/2}f^{7/10}\right) = 0$$

Integrating and using imposing a solution where $f \rightarrow \text{constant}$ as $\xi \rightarrow 0$, we find,

$$3\xi^2 \frac{df}{d\xi} + \xi^{7/2} f^{7/10} = 0$$

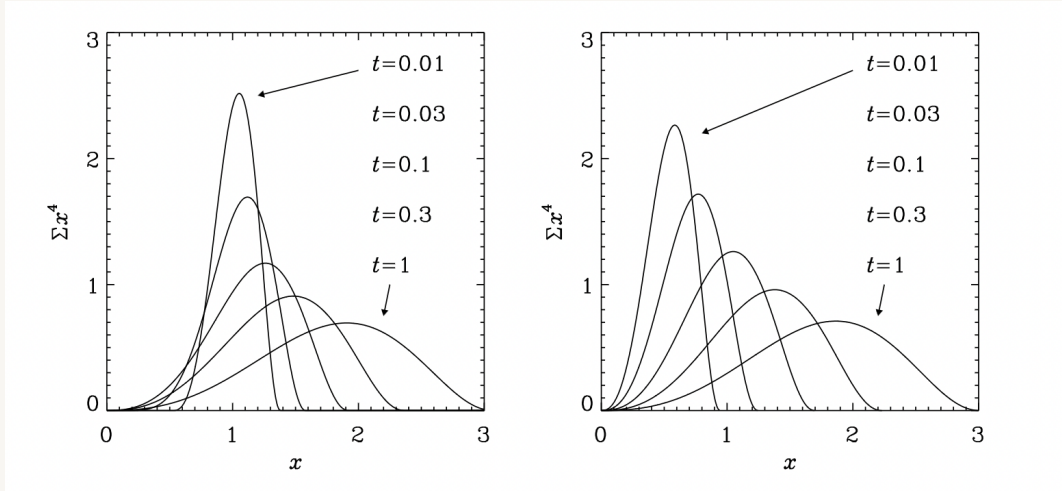
which we can integrate again to find,

$$f^{3/10} = \frac{1}{25}(\xi_{\text{out}}^{5/2} - \xi^{5/2})$$

where the constant of integration is taken as ξ_{out} . In fact, this value has physical significance and is the edge of our solution. This can be found by using the above relation and our relation for Σ and substituting this back into the conserved quantity C . After various cancellations, we are left with,

$$\int_0^{\xi_{\text{out}}} f^{7/10} \xi^{5/2} d\xi = 1$$

which by a numerical calculation, gives $\xi_{\text{out}} \approx 3.018$.



Here we see two solutions of the non-linear diffusion equation, in units such that $A = 1$. The left-hand panel shows the evolution from a narrow ring of unit radius. The right-hand panel shows the attracting similarity solution for the same total angular momentum, $C = 1$. The plotted variables show the redistribution of angular momentum. These are similarity solutions as they keep the same shape.

1.6 Vertical disc structure

1.6.1 Vertical hydrostatic equilibrium

The z -component of the conservation of motion (also known as the equation of motion for the disc) reads,

$$\rho \left(\frac{Du_z}{Dt} \right) = -\frac{\partial p}{\partial z} + \rho \frac{\partial \Phi}{\partial z} + \left(\frac{1}{r} \frac{\partial(rT_{rz})}{\partial r} + \frac{1}{r} \frac{\partial T_{\phi z}}{\partial \phi} + \frac{\partial T_{zz}}{\partial z} \right)$$

For a thin disc in axisymmetric motion, $\mathbf{u} = (0, r\Omega, 0)$ in the cylindrical coordinate basis and $\frac{\partial}{\partial \phi} = 0$. Recall that for such a disc, the only non-zero components of the stress tensor $\mathbf{T}$ are,

$$T_{r\phi} = T_{\phi r} = \mu r \frac{d\Omega}{dr}$$

which gives the z -component of the equation of motion as,

$$0 = -\frac{\partial p}{\partial z} + \rho \frac{\partial \Phi}{\partial z} \quad (7)$$

Expanding the potential about the mid-plane $z = 0$ (where the centre mass is located),

$$\Phi = \Phi(r, z) = \Phi(r, 0) + \Phi_{,z}(r, 0)z + \frac{1}{2}\Phi_{,zz}(r, 0)z^2 + \mathcal{O}(z^2)$$

where we used the axial-symmetry of the potential $\Phi = \Phi(r, z)$. Moreover, since we impose that the potential is symmetric in the plane $z = 0$, that is $\Phi(r, z) = \Phi(r, -z)$, we have that

$$\Phi_{,z}(r, z) = \frac{\partial \Phi(r, z)}{\partial z} = \frac{\partial \Phi(r, -z)}{\partial z} = -\frac{\partial \Phi(r, -z)}{\partial z} = -\frac{\partial \Phi(r, z)}{\partial z} = -\Phi_{,z}(r, z) \Rightarrow \Phi_{,z}(r, z) = 0$$

which gives,

$$\Phi = \Phi(r, z) = \Phi(r, 0) + \frac{1}{2}\Phi_{,zz}(r, 0)z^2 + \mathcal{O}(z^2)$$

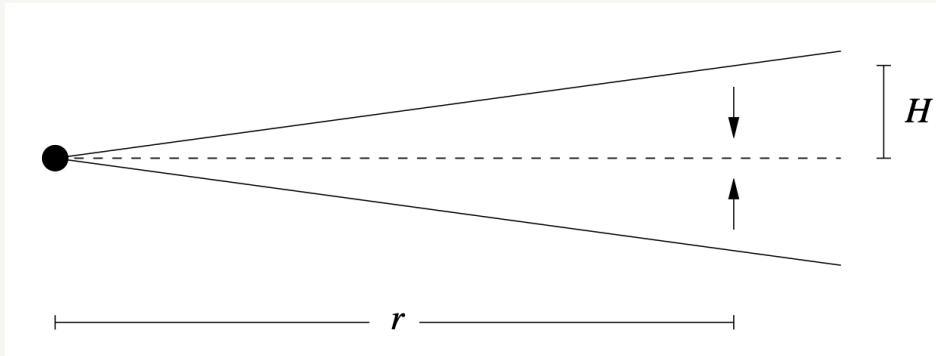
and recalling the vertical frequency $\Omega_z^2(r) = \Phi_{,zz}(r, 0)$ gives us,

$$\Phi = \Phi(r, 0) + \frac{1}{2}\Phi_{,zz}(r, 0)z^2 + \mathcal{O}(z^2) \Rightarrow \frac{\partial \Phi}{\partial z} = \Phi_{,zz}(r, 0)z + \mathcal{O}(z^2) \Rightarrow \frac{\partial \Phi}{\partial z} \approx \Omega_z^2 z$$

Additionally, in a Keplerian disc we previously found that $\Omega_z = \Omega$. Now, to leading order in z , the z -component of the equation of motion (7) reads,

$$\frac{\partial p}{\partial z} = -\rho \Omega_z^2 z \quad (8)$$

which is also known as the [equation of vertical hydrostatic equilibrium](#).



In the vertical direction, the pressure supports the disc against gravity.

Using approximations and simple scaling relations where we ignore any constants that are approximately equal to one are near essential in the following analysis.

An important parameter is the [angular semi-thickness](#) of the disc, H/r , where H represents the true semi-thickness, the height above the mid-plane or the density scale height. For a thin disc,

$$\frac{H}{r} \ll 1$$

From the equation of vertical hydrostatic equilibrium (8),

$$\frac{p}{H} \sim \rho \Omega^2 H$$

or upon introducing the [isothermal sound speed](#) $c_s = (p/\rho)^{1/2}$,

$$c_s \sim \Omega H$$

Observe that the dynamic viscosity has the same units as the pressure divided by the angular velocity, $[\mu] = [p/\Omega]$. This allows us to write,

$$\mu = \frac{\alpha p}{\Omega}$$

where α is a dimensionless viscosity parameter, and this is known as the [alpha viscosity prescription](#). It follows that,

$$\bar{\nu} \sim \frac{\mu}{\rho} \sim \frac{\alpha c_s^2}{\Omega} \sim \alpha c_s H$$

In kinetic theory, the kinematic viscosity is given by the relation

$$\nu \sim v \ell$$

where v is the mean speed of the molecules and ℓ is the mean free path. This is negligible for astrophysical discs. However, this can be used to form a similar estimate for the effective ‘eddy viscosity’ of the turbulence of the disc. This can be achieved by treating v as the mean turbulent speed and ℓ as the correlation length of the turbulence. For subsonic turbulence, $v \lesssim c_s$, and for eddies no longer than the disc thickness, $\ell \lesssim H$ and thus $\alpha \lesssim 1$. The stress in a Keplerian disc is therefore,

$$T_{r\phi} = \mu r \frac{d\Omega}{dr} = \frac{\alpha p}{\Omega} r \frac{d\Omega}{dr} = -\frac{3}{2} \alpha p$$

as for a Keplerian disc, $\Omega \sim r^{-3/2} \Rightarrow \frac{d\Omega}{dr} = -\frac{3}{2} \frac{\Omega}{r}$. The original paper gives $T_{r\phi} = -\alpha p$ as an alternative version of the alpha prescription.

We will now define the three important time scales: the dynamical, viscous and thermal time scales.

The **dynamical** time scale is the time scale of orbital motion and also vertical hydrostatic equilibrium,

$$t_{\text{dyn}} \sim \frac{1}{\Omega} \sim \frac{H}{c_s}$$

The **viscous** time scale is the time scale of the radial motion in the disc,

$$t_{\text{visc}} \sim \frac{r^2}{\bar{\nu}} \sim \alpha^{-1} \left(\frac{H}{r} \right)^{-2} t_{\text{dyn}}$$

Finally, the **thermal** time scale is the time scale for establishing the vertical thermal balance,

$$t_{\text{th}} \sim \frac{pH}{\bar{\nu}\Sigma\Omega^2} \sim \frac{c_s^2}{\bar{\nu}\Omega} \sim \frac{H^2}{\bar{\nu}} \sim \alpha^{-1} t_{\text{dyn}}$$

The **Mach number** of the orbital motion is given by,

$$\text{Ma} \sim \frac{r\Omega}{c_s} \sim \left(\frac{H}{r} \right)^{-1}$$

For a thin disc with $\alpha < 1$, we have,

$$t_{\text{dyn}} < t_{\text{th}} \ll t_{\text{visc}}$$

The characteristic radial velocity due to viscosity is,

$$|\bar{u}_r| \sim \frac{\bar{\nu}}{r} \sim \alpha \left(\frac{H}{r} \right) c_s$$

and for a thin disc $|\bar{u}_r| \ll c_s \ll r\Omega$, that is the orbital motion is highly supersonic while the accretion flow is highly subsonic.

Lastly, we will consider the relative contribution of the radial pressure gradient to the radial component of the equation of motion,

$$\frac{\partial p}{\partial r} \bigg/ \rho r \Omega^2 \sim \frac{c_s^2}{r^2 \Omega^2} \sim \left(\frac{H}{r} \right)^2$$

Vertical variations of the radial gravitational acceleration are also of this order. Other terms in the radial equation of motion, such as inertial terms associated with the radial motion, are smaller still. Hence,

$$u_\phi = r\Omega \left[1 + \mathcal{O} \left(\left[\frac{H}{r} \right]^2 \right) \right]$$

which shows that treating the azimuthal fluid velocity as equal to the orbital velocity of a test particle is an excellent approximation for a thin disc.

2 Example Sheet 1

2.1 Revision of Keplerian orbits

The equation of motion of a test particle in the gravitational field of a point mass M is

$$\ddot{\mathbf{r}} = -\frac{GM}{|\mathbf{r}|^3}\mathbf{r}$$

where $\mathbf{r}$ is the position of the test particle.

Observe that if the position of the particle is confined to a plane containing the central mass, the normal vector of that plane $\mathbf{n}$ would be given by

$$\mathbf{n} = \mathbf{r} \times \dot{\mathbf{r}}$$

due to the position $\mathbf{r}$ and the velocity $\dot{\mathbf{r}}$ lying in the plane⁴. Moreover, the quantities $\mathbf{r} = \mathbf{r}(t)$ and $\dot{\mathbf{r}} = \dot{\mathbf{r}}(t)$ are functions of time, yet the normal vector of the plane is a constant vector. This results in the following fact.

$$\mathbf{r} \text{ is confined in the plane} \iff \frac{d}{dt}(\mathbf{r} \times \dot{\mathbf{r}}) = \mathbf{0}$$

Armed with this and the product rule, we can calculate,

$$\frac{d}{dt}(\mathbf{r} \times \dot{\mathbf{r}}) = \dot{\mathbf{r}} \times \dot{\mathbf{r}} + \mathbf{r} \times \ddot{\mathbf{r}} = \mathbf{r} \times \left(-\frac{GM}{|\mathbf{r}|^3}\mathbf{r}\right) = -\frac{GM}{|\mathbf{r}|^3}(\mathbf{r} \times \mathbf{r}) = \mathbf{0}$$

where we have used the anti-commutativity property of the cross product ($\mathbf{a} \times \mathbf{a} = \mathbf{0}$) and the fact that $|\mathbf{r}|$ is a scalar. Thus, the motion of the test particle is confined to a plane containing the central mass⁵.

Let (r, ϕ) be polar coordinates in the above plane, that is,

$$\mathbf{r} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r \cos \phi \\ r \sin \phi \end{pmatrix}$$

and that $|\mathbf{r}| = r$. Using the product and the chain rule,

$$\dot{\mathbf{r}} = \begin{pmatrix} \dot{r} \cos \phi - r \dot{\phi} \sin \phi \\ \dot{r} \sin \phi + r \dot{\phi} \cos \phi \end{pmatrix} \quad \text{and} \quad \ddot{\mathbf{r}} = \begin{pmatrix} \ddot{r} \cos \phi - 2\dot{r}\dot{\phi} \sin \phi - r\ddot{\phi} \sin \phi - r\dot{\phi}^2 \cos \phi \\ \ddot{r} \sin \phi + 2\dot{r}\dot{\phi} \cos \phi + r\ddot{\phi} \cos \phi - r\dot{\phi}^2 \sin \phi \end{pmatrix}$$

⁴If this was not true then the particle would fly out of the supposed plane!

⁵You might be (correctly) suspicious about the argument since we could have that $\mathbf{n}$ is zero. A bit more thinking shows that this would correspond to $\mathbf{r} \parallel \dot{\mathbf{r}}$ which means that the motion is in fact constricted to a straight line and you can take any plane containing that straight line.

which we can use in our equation of motion to find,

$$\begin{pmatrix} \ddot{r} \cos \phi - 2\dot{r}\dot{\phi} \sin \phi - r\ddot{\phi} \sin \phi - r\dot{\phi}^2 \cos \phi \\ \ddot{r} \sin \phi + 2\dot{r}\dot{\phi} \cos \phi + r\ddot{\phi} \cos \phi - r\dot{\phi}^2 \sin \phi \end{pmatrix} = -\frac{GM}{r^3} \begin{pmatrix} r \cos \phi \\ r \sin \phi \end{pmatrix}$$

Aiming to use our constraint of $\frac{d}{dt}(\mathbf{r} \times \dot{\mathbf{r}}) = \mathbf{0}$, we find that,

$$\mathbf{r} \times \dot{\mathbf{r}} = \det \begin{pmatrix} r \cos \phi & r \sin \phi \\ \dot{r} \cos \phi - r\dot{\phi} \sin \phi & \dot{r} \sin \phi + r\dot{\phi} \cos \phi \end{pmatrix} = r^2 \dot{\phi} (\sin^2 \phi + \cos^2 \phi) = r^2 \dot{\phi}$$

which gives us

$$\frac{d}{dt}(r^2 \dot{\phi}) = 0 \Rightarrow \boxed{r^2 \dot{\phi} = h = \text{constant}}$$

Observe that our constraint further tells us that,

$$\frac{d}{dt}(r^2 \dot{\phi}) = r(2\dot{r}\dot{\phi} + r\ddot{\phi}) = 0$$

which we can use to simplify our equations as,

$$\begin{aligned} & \begin{pmatrix} \ddot{r} \cos \phi - \sin \phi (2\dot{r}\dot{\phi} + r\ddot{\phi}) - r\dot{\phi}^2 \cos \phi \\ \ddot{r} \sin \phi + \cos \phi (2\dot{r}\dot{\phi} + r\ddot{\phi}) - r\dot{\phi}^2 \sin \phi \end{pmatrix} = -\frac{GM}{r^3} \begin{pmatrix} r \cos \phi \\ r \sin \phi \end{pmatrix} \\ \Rightarrow & \begin{pmatrix} \ddot{r} \cos \phi - r\dot{\phi}^2 \cos \phi \\ \ddot{r} \sin \phi - r\dot{\phi}^2 \sin \phi \end{pmatrix} = -\frac{GM}{r^2} \begin{pmatrix} \cos \phi \\ \sin \phi \end{pmatrix} \\ \Rightarrow & \begin{pmatrix} \ddot{r} - r\dot{\phi}^2 \\ \ddot{r} - r\dot{\phi}^2 \end{pmatrix} = -\frac{GM}{r^2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow \boxed{\ddot{r} - r\dot{\phi}^2 = -\frac{GM}{r^2}} \end{aligned}$$

Using our constraint equation $r^2 \dot{\phi} = h$, we can write,

$$\ddot{r} - \frac{h^2}{r^3} = -\frac{GM}{r^2}$$

Letting $r(\phi) = \frac{1}{s(\phi)}$, this relationship gives,

$$\begin{aligned} \dot{r} &= -\frac{1}{s^2} \frac{ds}{d\phi} \dot{\phi} = -h \frac{ds}{d\phi} = -hs', \\ \ddot{r} &= -h^2 s^2 s'', \end{aligned}$$

where we are using the notation $s' = \frac{ds}{d\phi}$ and $s'' = \frac{d^2 s}{d\phi^2}$. Consequently,

$$s'' + s = \frac{GM}{h^2}$$

which we can solve with a homogeneous and inhomogeneous part to give

$$s(\phi) = \frac{GM}{h^2} + A \cos(\phi - \omega)$$

and solving for r gives,

$$r(\phi) = \frac{\lambda}{1 + e \cos(\phi - \omega)}$$

where $\lambda = \frac{h^2}{GM}$.

The orbits for the cases $e = 0, 0 < e < 1, e = 1$ and $e > 1$ correspond to circular, elliptic, parabolic and hyperbolic orbits respectively.

Finally, the total energy of the test particle is given by,

$$E = T + V = \frac{1}{2}(\dot{r}^2 + r^2\dot{\phi}^2) - \frac{GM}{r}$$

where we used that the kinetic energy in polar coordinates is $\frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2)$. Given angular momentum h , we can write this as,

$$E = \frac{1}{2} \left(\dot{r}^2 + \frac{h^2}{r^2} \right) - \frac{GM}{r}$$

Observe that the kinetic energy is minimal when $\dot{r} = 0$ and the potential energy is always negative. When $\dot{r} = 0$, the total energy is,

$$E(r) = \frac{h^2}{2r^2} - \frac{GM}{r}$$

To find the orbit with least energy, we can differentiate,

$$\frac{dE}{dr} = -\frac{h^2}{r^3} + \frac{GM}{r^2} = 0 \Rightarrow r = \frac{h^2}{GM} = \lambda$$

which gives the circular orbit of radius λ .